TEXORA · LECTURE SHEET

Introduction of Man-Made Fibre

Fibres made by industry rather than grown: their definition and classification, the history from Chardonnet's artificial silk to the synthetic age, and how they differ from the natural fibres they joined.

TE-2108 · MAN-MADE TEXTILE FIBERS · TOPIC 1 · LEVEL 2 TERM 1

1. Definition and classification

A **man-made fibre** (also "manufactured fibre") is a fibre produced industrially by extruding a fibre-forming substance through fine holes (a **spinneret**) and solidifying the streams into continuous filaments — as opposed to natural fibres, which grow as fibres in plants and animals. Two great families arise from the starting material:

- **Regenerated fibres** — the polymer itself is natural (cellulose from wood pulp or cotton linters, or protein), dissolved and re-formed into fibre: viscose, lyocell, modal, cellulose acetate. The polymer chains already exist; industry only reshapes them (and, in acetate, chemically modifies them).
- **Synthetic fibres** — the polymer is first built from small petrochemical monomers by polymerisation (TE-2107), then spun: polyester, nylon, acrylic, polypropylene, spandex.

A few fibres stand outside both families: **inorganic** man-made fibres such as glass fibre and carbon fibre, drawn or derived from non-polymeric feedstocks.

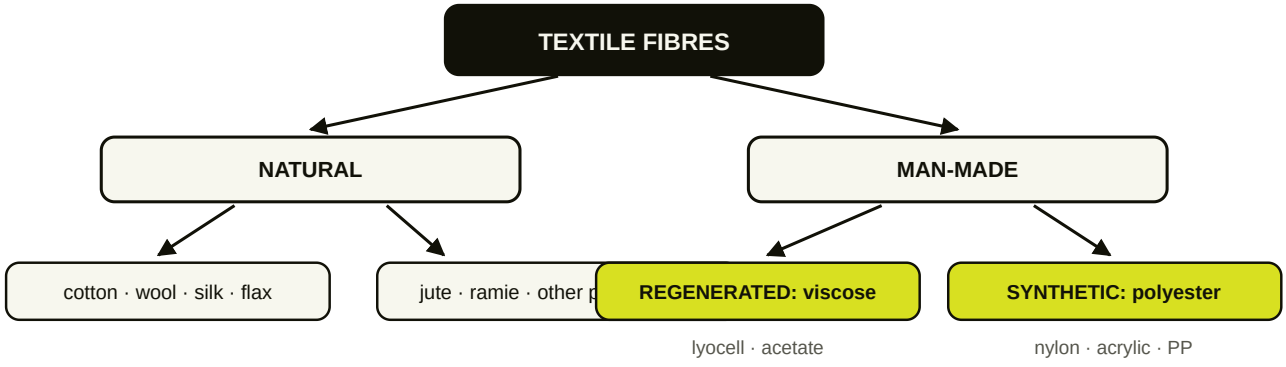


Figure 1 — The fibre family tree. Man-made fibres divide by feedstock: regenerated fibres reshape a natural polymer; synthetic fibres first build the polymer from monomers.

2. A short history

YEAR	MILESTONE
1664	Robert Hooke predicts artificial silk from a gummy fluid, two centuries early
1884–1891	Count Hilaire de Chardonnet patents and produces "artificial silk" from nitrocellulose — beautiful but dangerously flammable; the first commercial man-made fibre
1892	Cross, Bevan and Beadle patent the viscose process in England
1905	Courtaulds begins commercial viscose rayon production — cheap, silk-like, and safe enough to found an industry
1920s	Cellulose acetate fibres commercialised; "rayon" adopted as the generic name (1924)
1935–1939	Carothers at DuPont invents nylon; first nylon stockings sold 1940 — the first truly synthetic fibre, from coal/air/water feedstocks via petrochemicals
1941	Whinfield and Dickson (UK) patent polyester (PET); commercialised in the 1950s as Terylene/Dacron
1950s	Acrylic fibres (Orlon, Acrilan) commercialised; Natta's isotactic polypropylene (1954–57) opens the polyolefin fibre era
1959	Spandex (elastane) introduced — the first fibre designed for stretch
1980s–90s	Microfibres, high-performance aramids spread; lyocell (solvent-spun cellulose) launched as the environmentally improved regenerated fibre

Why did industry bother? Silk was beautiful but scarce and expensive; cotton and wool output is capped by land, season and animal. Man-made fibres offered **unlimited, uniform, tailorable** supply: length, fineness, cross-section, lustre and even chemistry could be specified by the producer instead of negotiated with nature.

3. Forms in which man-made fibres are supplied

- **Filament yarn** — one (monofilament) or many (multifilament) continuous filaments, used as-spun: hosiery, linings, tyre cord.
- **Staple fibre** — filaments cut to cotton- or wool-like lengths (typically 32–150 mm) for spinning on natural-fibre systems, alone or in blends.
- **Tow** — a huge bundle of parallel filaments (hundreds of thousands) sold for cutting or stretch-breaking into staple/top.
- **POY, FDY, DTY** — partially oriented, fully drawn and draw-textured yarns: the stages of orientation and bulk at which polyester/nylon filament is sold.

Fineness is specified in **denier** (grams per 9000 m) or **dtex** (grams per 10,000 m), usually as total yarn count and filaments: 150 denier/48 filaments means each filament is 150 ÷ 48 = **3.125 denier** (denier per filament, dpf). Filaments finer than about 1 dpf are called **microfibres**.

4. Man-made versus natural fibres

FEATURE	NATURAL FIBRES	MAN-MADE FIBRES
Length	Fixed by growth (cotton 20–50 mm; wool varies)	Any length — filament or chosen staple
Uniformity	Varies with season, plant, animal	Controlled to specification, batch after batch
Fineness & cross-section	Given by nature (cotton ribbon, wool round)	Designed: round, trilobal, hollow, multilobal — section shapes gloss, warmth, soil-hiding
Strength & durability	Moderate (cotton, wool); silk high	Adjustable; synthetics generally stronger and more abrasion-resistant
Moisture behaviour	High regain, comfortable, slow drying	Regenerated: high regain; synthetics: low regain, fast drying, static-prone
Heat behaviour	Char, do not melt	Most synthetics melt — iron carefully; regenerated char like cotton
Supply	Agricultural limits, price swings	Industrial, scalable; price follows petrochemical feedstocks (except regenerated)
End of life	Biodegradable	Regenerated cellulose biodegrades; synthetics persist (TE-2107 Topic 6)

5. What makes a polymer fibre-forming?

Not every polymer can be a fibre. The requirements, which organise the rest of this course: (i) **linear chains** of sufficient DP (≥ about 100) that can be aligned; (ii) **regular structure** able to crystallise on drawing; (iii) **inter-chain forces** strong enough for a useful melting/softening point but processable — the polymer must melt without degrading, or dissolve in a practical solvent; (iv) **stability** to water, light, washing and wear in service. Polyethylene fails (iii)/(iv) as apparel — too low a melting point and no dye sites; PVC fails on heat stability; that is why the commercial fibre list (Topics 2–4) is short although the polymer list (TE-2107 Topic 3) is long.

Glossary

Spinneret — the metal plate (or cap) pierced with fine holes through which spinning fluid is extruded. **Rayon** — the generic name for regenerated cellulose fibres (viscose, lyocell, modal). **Tow** — a large parallel bundle of filaments destined for cutting into staple. **dpf** — denier per filament, total denier divided by filament count. **Microfibre** — a filament finer than about 1 denier per filament (≈ 1.1 dtex per filament). **POY/DTY** — partially oriented yarn / draw-textured yarn, intermediate filament products.

Key points

• Man-made = spinneret-formed: regenerated fibres reshape natural polymer, synthetic fibres polymerise first.
• History runs nitrocellulose (1891) → viscose (1905) → nylon (1939) → polyester, acrylic, PP (1950s) → lyocell and microfibres.
• The industrial advantages are uniformity and design freedom — length, fineness, cross-section, lustre are specified, not harvested.
• Fibre-forming polymers must be linear, crystallisable, of adequate DP, and meltable or soluble without degrading.

Practice

1. A polyester yarn is labelled 167 dtex/48 filaments. Calculate the dtex per filament and say whether it counts as microfibre (microfibre = below about 1 denier ≈ 1.1 dtex per filament).
2. Classify each fibre as natural, regenerated or synthetic, and name its polymer: (a) modal; (b) elastane; (c) ramie; (d) acetate; (e) aramid.
3. Why did viscose, not Chardonnet's nitrocellulose fibre, found the man-made fibre industry?
4. Give three properties a garment maker can specify for a man-made fibre that cannot be specified for cotton.
5. Polyethylene is the world's largest-volume polymer but a minor apparel fibre. Which fibre-forming requirements does it fail?

Answer key

1. 167 ÷ 48 = **3.48 dtex per filament** — well above the microfibre boundary; it is a normal textile filament.
2. (a) Regenerated, cellulose. (b) Synthetic, segmented polyurethane. (c) Natural (plant bast), cellulose. (d) Regenerated (modified), cellulose acetate. (e) Synthetic, aromatic polyamide.

3. Nitrocellulose fibre was dangerously flammable (its polymer is a close relative of gun-cotton), costly in solvents, and required a denitration step to be safe at all. Viscose used cheap wood pulp, water-based chemistry after the xanthate step, and produced a safe cellulose fibre at a price that built factories and markets.
4. Any three: filament length/staple length; fineness (denier per filament); cross-sectional shape (trilobal, hollow); lustre grade (TiO_2 level); crimp level; dye-affinity modification; flame-retardant comonomer content.
5. It fails on inter-chain forces and service temperature: only van der Waals cohesion gives a melting point too low for ironing and hot washing, no polar groups means no dyeability, and it creeps under load. It succeeds only where those do not matter (ropes, nets, geotextiles).
